

Early menarche and childbirth accelerate aging-related outcomes and age-related diseases: Evidence for antagonistic pleiotropy in humans

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eLife Assessment

This **important** study uses Mendelian Randomisation to show that early life phenotypes (i.e. onset of age at menarche and age at first birth) have an influence on a multitude of health outcomes later in life. The provided empirical evidence supporting the antagonistic pleiotropy theory is **solid**. However, some additional analyses and a more comprehensive discussion of the findings are needed to make the study stronger.

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Abstract

Aging can be understood as a consequence of the declining force of natural selection with age. Consistent with this the antagonistic pleiotropic theory of aging suggests that aging results from the trade-offs that promote early growth and reproduction. However, evidence for antagonistic pleiotropy in humans is largely lacking. Using Mendelian Randomization (MR), we demonstrated that later ages of menarche or first childbirth were genetically associated with longer parental lifespan, decreased frailty index, slower epigenetic aging, later menopause, and reduced facial aging. Moreover, later menarche or first childbirth were also genetically associated with a lower risk of several age-related diseases, including late-onset Alzheimer's disease (LOAD), type 2 diabetes, heart disease, essential hypertension, and chronic obstructive pulmonary disease (COPD). We validated the associations between the age of menarche, childbirth, and the number of childbirths with several age-related outcomes in the UK Biobank by conducting regression analysis of nearly 200,000 subjects. Our results demonstrated that menarche before the age 11 and childbirth before 21 significantly accelerated the risk of several diseases, and almost doubled the risk for diabetes, heart failure, and quadrupled the risk of obesity, supporting the antagonistic pleiotropy theory. We identified 128 significant single nucleotide polymorphisms (SNPs) that influenced age-related outcomes, some of which were involved in known longevity pathways, including IGF1, growth hormone, AMPK, and mTOR signaling. Our study also identified higher BMI as a mediating factor in causing the increased risk of certain diseases, such as type 2 diabetes and heart failure, in women with early menarche or early pregnancy, emphasizing the

importance of the thrifty gene hypothesis in explaining in part the mechanisms behind antagonistic pleiotropy. Our study highlights the complex relationship between genetic legacies and modern diseases, emphasizing the need for gender-sensitive healthcare strategies that consider the unique connections between female reproductive health and aging.

Introduction

Given the decline in the force of natural selection with age, genes that influence aging and age-related diseases are likely to be selected for their influence on early life events ¹. The theory of antagonistic pleiotropy posits that inherent trade-off in natural selection, where genes beneficial for early survival and reproduction may have costly consequences later, contributes to the aging process and age-related diseases ¹⁻³. Though evidence for antagonistic pleiotropy has been observed in invertebrate models, evidence for causal relationships in mammals, especially humans, is largely lacking ⁴. The timing of reproductive events, such as menarche and childbirth, has long been recognized as a crucial aspect of human life history and evolution. However, accumulating evidence suggests that these reproductive milestones may have far-reaching implications ^{5,6}.

Considering reproductive events are largely regulated by genetic factors that can manifest the physiological outcome later in life, we hypothesized that earlier or later onset of menarche and first childbirth could reflect broader genetic influences on longevity and disease susceptibility, serving as proxies for biological aging processes (fig. S1 and fig. S2). Two-sample and two-step Mendelian randomization (MR) analyses were adopted to explore the genetic causal associations using the inverse variance weighted (IVW) model. After preprocessing, there were 209 and 33 single nucleotide polymorphisms (SNPs) included in MR analyses for the two exposures, age at menarche ⁷ and age at first birth ⁸, respectively.

Later menarche was genetically associated with later aging outcomes

Since parental ages at death may reflect the genetic heritability of lifespan, parental ages at death were used as general aging outcomes ⁹. Compared to early menarche, later age at menarche was significantly associated with later parental ages at death for mothers or fathers (OR=1.023, $P=1.84\times 10^{-4}$; OR=1.020, $P=7.88\times 10^{-4}$ for father and mother's ages at death respectively) (Fig. 1A and Table S1-S3). Aging is the biggest risk factor for frailty and several age-related diseases. Later age at menarche was associated with a lower frailty index, which was calculated using a set of 49 self-reported questionnaire items on traits covering health, presence of diseases and disabilities, and mental well-being ¹⁰ (OR=0.977, $P=5.75\times 10^{-3}$). After excluding the instrumental variables (IVs) associated with body mass index (BMI), the significant association between later menarche and outcomes remained based on the weighted median model (WMM) for frailty index (OR=0.972, $P=0.0202$) and based on IVW for parental ages at death (OR=1.015, $P=2.46\times 10^{-2}$; OR=1.012, $P=3.87\times 10^{-2}$). Next, we examined the associations between the age of menarche and organ aging or age-related diseases (Fig. 1B). Later age of menarche was associated with later menopause onset (OR=1.121, $P=2.11\times 10^{-2}$ with WMM), lower risks of late-onset Alzheimer's disease (LOAD) (OR=0.897, $P=3.88\times 10^{-2}$), chronic heart failure (CHF) (OR=0.953, $P=2.80\times 10^{-2}$), essential hypertension (OR=0.990, $P=1.99\times 10^{-8}$), facial aging (OR=0.983, $P=2.16\times 10^{-9}$), and early onset chronic obstructive pulmonary disease (COPD) (OR=0.838, $P=9.50\times 10^{-4}$) and mildly higher risk of osteoporosis (OR=1.001, $P=2.07\times 10^{-2}$). These associations were still significant after excluding SNPs associated with BMI, except for CHF. One potential mechanism of antagonistic pleiotropy is accelerated cell growth and division; therefore, we examined the relationship between menarche

and certain cancers. Later age at menarche was associated with lower risks of breast cancer ($OR=0.859$, $P=1.53\times 10^{-2}$) and endometrial cancer ($OR=0.896$, $P=9.02\times 10^{-4}$) compared with early menarche (**Fig. 1B** [↗](#)).

Later age at first birth was genetically associated with later aging outcomes

Next, we examined the associations between the age of childbirth and age-related outcomes. Compared to early first birth, later age at first birth was associated with lower frailty index ($OR=0.937$, $P=5.93\times 10^{-11}$) and GrimAge, a measure of epigenetic aging ($OR=0.447$, $P=4.37\times 10^{-2}$), and older parental ages at death ($OR=1.035$, $P=6.88\times 10^{-5}$ for father's age at death; $OR=1.034$, $P=1.78\times 10^{-4}$ for mother's age at death) (**Fig. 1A** [↗](#)). Similar to results with the age of menarche, later age at first birth was associated with later menopause onset ($OR=1.306$, $P=3.08\times 10^{-4}$), lower risks of type 2 diabetes ($OR=0.890$, $P=3.65\times 10^{-6}$), CHF ($OR=0.926$, $P=2.56\times 10^{-2}$), essential hypertension ($OR=0.989$, $P=1.54\times 10^{-4}$), and gastrointestinal or abdominal disease (GAD) ($OR=0.991$, $P=1.29\times 10^{-3}$). Most significant associations remained after the BMI-related SNPs were excluded, except for CHF (**Fig. 1C** [↗](#) and Table S3). Furthermore, later age at first birth was also significantly associated with a lower risk of cervical cancer ($OR=0.999$, $P=1.52\times 10^{-2}$) but not breast and endometrial cancers (**Fig. 1C** [↗](#)).

BMI is an important mediator in significant associations

As BMI is an important modulator of aging ¹¹[↗](#), we examined its role in explaining these associations. Based on the two-sample MR analyses, significant associations between early menarche and CHF, and between early first birth and CHF and cervical cancer, were not observed after excluding SNPs related to BMI (**Fig. 1** [↗](#)). To estimate the effect of BMI as the mediator, we further conducted two-step MR. Exposures of age of menarche and age at first birth, and outcomes of frailty index, type 2 diabetes, and CHF, were included in the two-step MR analysis. Later ages of menarche and first birth were associated with lower BMI ($OR=0.948$, $P=2.73\times 10^{-13}$; $OR=0.956$, $P=6.85\times 10^{-5}$). Higher BMI was associated with higher risks of type 2 diabetes ($OR=2.391$, $P=4.65\times 10^{-156}$), CHF ($OR=1.690$, $P=1.67\times 10^{-44}$), and cervical cancer ($OR=1.001$, $P=3.16\times 10^{-2}$). Significant pleiotropy was detected in the association between BMI and frailty index. Two-step MR showed that BMI significantly mediated the associations between menarche and, type 2 diabetes or CHF (Table S4). BMI also significantly mediated the associations between first birth and, type 2 diabetes, CHF, or cervical cancer (Table S4). As both early menarche and first birth were associated with higher BMI, the BMI had a significant effect to explain the associations partially. Results of colocalization analysis showed 5 SNPs associated with both the exposures and outcomes (Table S5).

Mechanisms explaining antagonistic pleiotropy

We identified 128 significant SNPs in the genes from the MR analysis for their association between age of menarche or first birth with 15 aging and age-related disease outcomes. These SNPs were then listed according to (i) the number of study outcomes they were associated with and (ii) their frequency of occurrence in the European and global population. We chose to depict only the SNPs that were associated with greater than three outcomes as shown in **Fig. 2** [↗](#). The SNP rs2003476 in the *CRTC1* gene was found to be associated with seven aging and age-related disease outcomes (frailty index, LOAD, CHF, father's age at death, mother's age at death, breast, and endometrial cancer). Previous studies have indicated that *CRTC1* transcription domain is linked with extending the lifespan in *C. elegans* ¹²[↗](#) and a decrease in *CRTC1* levels was found to be associated with human AD ¹³[↗](#). Two other genes involved in the glutathione metabolism pathway, *CHAC1* and *GGT7*, were also found to be associated with four different aging outcomes (*CHAC1* was associated with frailty index, diabetes, COPD, and BMI; *GGT7* was associated with father's age at death, hypertension, breast cancer and BMI) (**Fig. 2** [↗](#)). Importantly, the SNP in *CHAC1* is a missense

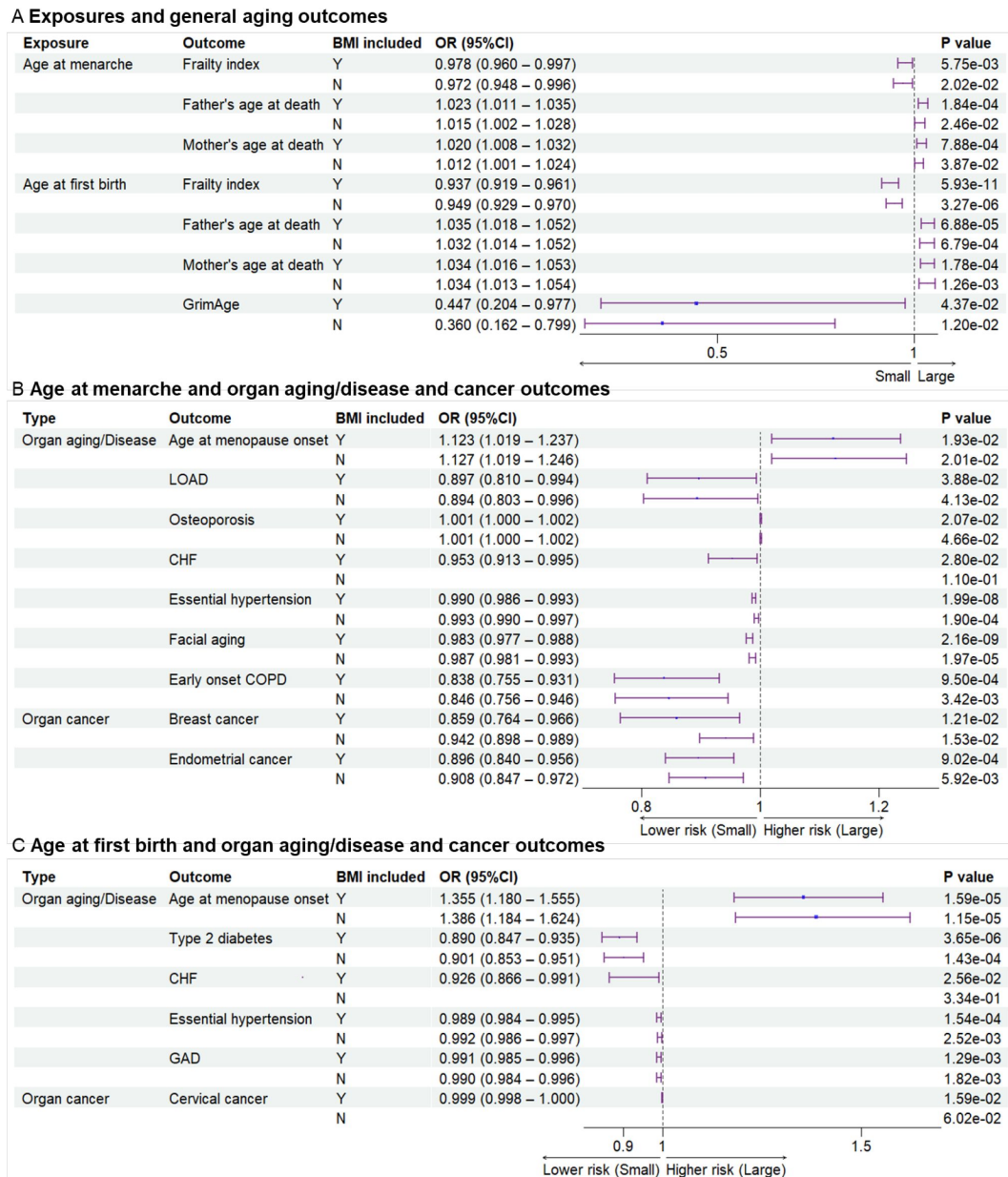


Figure 1.

Genetic associations between exposures and outcomes.

(A), genetic associations between exposures and general aging. Later age at menarche was associated with a lower frailty index and higher parental ages at death. Later age at first birth was associated with lower frailty index and GrimAge and higher parental ages at death. (B), genetic associations between age at menarche and outcomes of organ aging and diseases as well as organ cancer. Later age at menarche was associated with later age at menopause onset and lower risks of LOAD, CHF, essential hypertension, facial aging, early onset COPD, breast cancer, and endometrial cancer. (C), genetic associations between age at first birth and outcomes of organ aging and diseases as well as organ cancer. Later age at first birth was associated with later age at menopause onset and lower risks of type 2 diabetes, CHF, essential hypertension, gastrointestinal or abdominal disease, and cervical cancer. The significant associations were not detected after BMI-related SNPs excluded for outcomes of CHF and cervical cancer. BMI included, BMI-related SNPs included; CHF, chronic heart failure; GAD, gastrointestinal or abdominal disease.

variant, suggesting its impact on protein translation and potential contribution to disease biology. CHAC1 has recently been implicated in age-related macular degeneration¹⁴ and muscle wasting¹⁵. Posttranslational protein-lipid modification processes are known to contribute to aging and age-related diseases¹⁶. N-myristoylation is one such process that is catalyzed by NMT (N-myristoyltransferase)¹⁷, another intriguing gene candidate identified in our analysis. We found an association of *NMT1* with three aging outcomes: T2D, CHF, and endometrial cancer. Various myristoylated proteins have been implicated in diverse intracellular signaling pathways¹⁸ including AMP-activated protein kinase (AMPK)^{19,20}. It has been shown that NMT1 exerts synovial tissue-protective functions by promoting the recruitment of AMPK to lysosomes and inhibiting mTORC1 signaling¹⁹. In addition, drugs targeting NMTs (NMT1 and NMT2) have been suggested to be potent senolytics²¹ and a target for treating or preventing various diseases such as cancer^{22,23} and heart failure²⁴. Furthermore, modulating the evolutionarily conserved N-myristoyl transferase NMT1 was shown to extend lifespan in yeast²⁵. The full table of SNPs and their population frequency and association with aging outcomes is provided in Table S6.

Canonical signaling pathways analysis

A total of 498 canonical signaling pathways related to 128 SNPs/gene outcomes were identified based on the Ingenuity Pathway Knowledge Base (IPKB). After ranking the identified canonical signaling pathways according to their adjusted P-values, the top 25 canonical signaling pathways with a p-value < 10^{-2} enriched by SNPs/genes involved in the age of menarche and first birth are represented in **Fig. 3A**. The top enriched canonical signaling pathways fell into these broader categories: (1) Developmental and cellular signaling pathways such as BMP, IGF-1, and growth hormone signaling; (2) Neuronal signaling and neurological disorders that include amyloid processing, activation of NMDA receptors and postsynaptic events, and glioma signaling; (3) Metabolic signaling pathways such as insulin receptor, mTOR, and leptin signaling; (4) Immunological and inflammatory signaling pathways, which include lymphotoxin β receptor and TREM1 signaling; (5) Cardiovascular and muscular signaling pathways including the role of NFAT in cardiac hypertrophy. In addition to canonical pathways, SNPs/genes were shown to be further associated with diseases and functions. IPA showed a total of 80 diseases and functions associated with age at menarche and first birth. These associated diseases and functions were rated according to their adjusted P-values, and the top 20 enriched categories with a P-value < 10^{-5} were selected and depicted in **Fig. 3B**.

Gene network analysis and drug interactions

Besides pathway and disease associations, the 128 SNPs/genes were further connected to identify 11 functional networks using IPA. These IPA networks were ranked based on their consistency scores. The top 5 networks included a range of 13-17 genes with scores above 25, indicating robust regulatory analysis²⁶ (Table S7). The top networks were mainly connected to the following functions: Cellular interaction/signaling/development (**Fig. 3C**), cell death and survival, cancer, gastrointestinal, reproductive (**Fig. 3C**), neurological (**Fig. 3D**), and cardiovascular system, development, function, diseases. These results support the notion that dynamic cellular interactions and the development of various vital organ systems influenced by age at menarche and first birth also influence age-related diseases. The two top IPA networks were further outlined in **Fig. 3C and 3D**. To further explore the drugs targeting genes associated with age at menarche and first birth, we conducted an analysis using ChEMBL²⁷ and DrugBank databases²⁸. This examination revealed that a total of 11 FDA-approved drugs target the identified genes (*PRKAG2*, *SCN2A*, *DPYD*, *MC3R*, *AKT*, *MAPK*, *KCNK9*, *TACR3*, *HTR4*, *KCNB2*, *RXRG*) (Table S8).

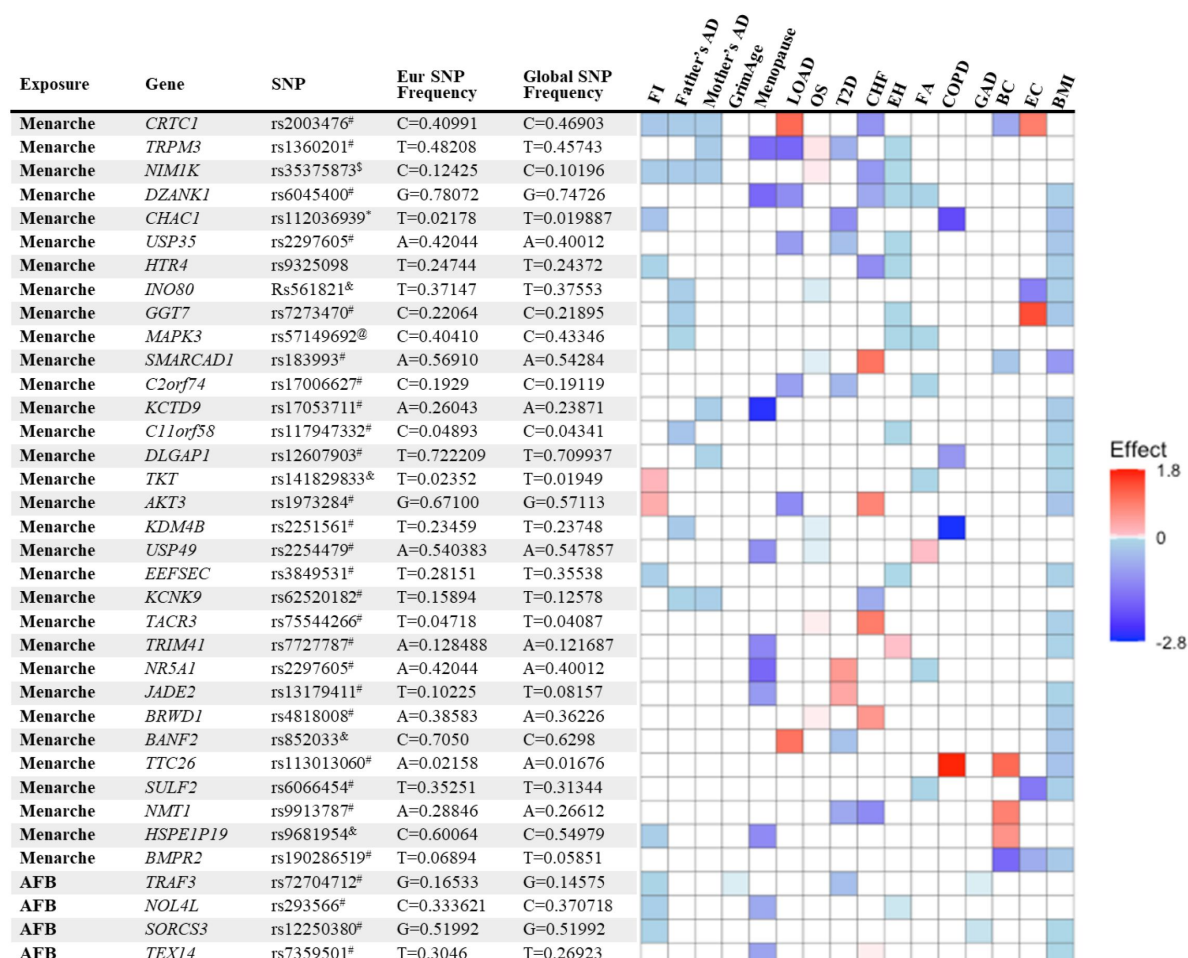


Figure 2.

List and heatmap of SNPs/genes from post-MR analysis showing an association between age at menarche and first birth with three or more aging outcomes.

Genomic region location for each variant is depicted by different characters: intron variant[#]; 2KB upstream variant; Missense variant^{*}; Intergenic variant[&]; Regulatory region variant[@]. Heatmap is representative of an association of each gene/SNP with different aging outcomes. Red bar represents a harmful association (-beta for Father's DA, Mother's DA, and menopause; +beta for other outcomes); Blue bar represents a beneficial association (+beta for Father's DA, Mother's DA, and menopause; -beta for other outcomes); White bar represents no association. AFB; age at first birth; FI, frailty index; Father's DA, father's age at death; Mother's DA, mother's age at death; Menopause, age at menopause; LOAD, late-onset Alzheimer's disease; OS, osteoporosis; T2D, type II diabetes; CHF, chronic heart failure; EH, essential hypertension; FA, facial aging; COPD, chronic obstructive pulmonary disease; GAD, gastrointestinal or abdominal disease; BC, breast cancer; EC, endometrial cancer; BMI, body mass index.

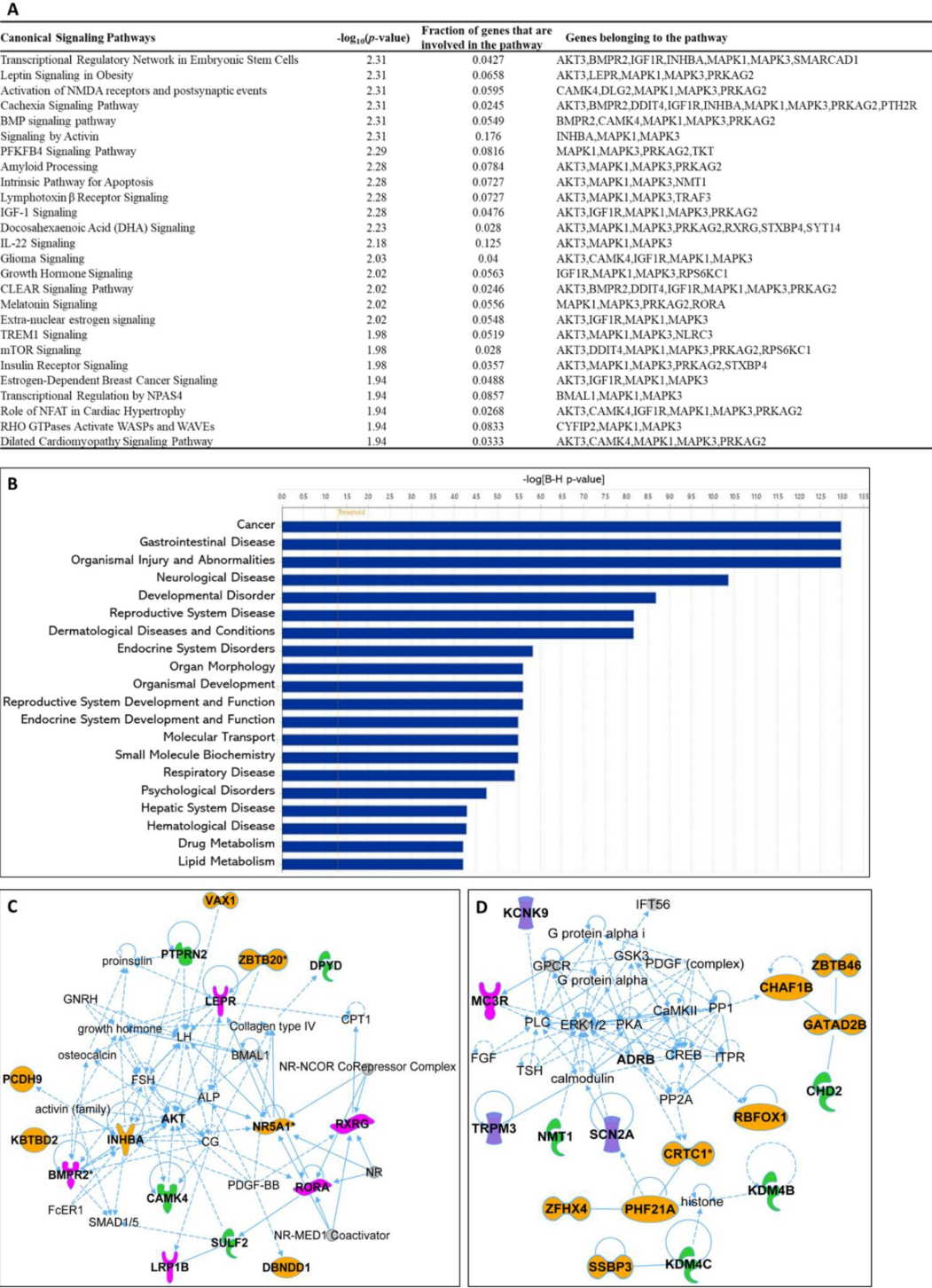


Figure 3.
Ingenuity Pathway Analysis (IPA). SNPs/gene outcomes from MR analysis were subjected to IPA.

(A) Canonical signaling pathways in age at menarche and first birth. The adjusted P-values of the top 25 signaling pathways are listed. (B) Disease and functions in age at first birth and menarche. The adjusted P-value of the top 20 significantly involved diseases and functions are listed. (C) IPA network 1 and (D) IPA network 2. The genes from our post-MR analysis are in bold. Solid lines indicate direct connections, while dotted lines indicate indirect connections (circular arrows mean influence itself). Color coding: pink-receptors, green-enzymes, purple-channels, orange-other proteins.

Population validation for associations between age at menarche, or first live birth, or number of births with age-related outcomes

Next, we validated the genetic associations in the population cohort of the UK Biobank. 264,335, 184,481, and 259,758 participants were included for independent variables of age at menarche, age at first live birth, and number of births, respectively (Table S9). Ages at menarche were divided into 5 age groups, <11y, 11-12y, 13-14y, 15-16y, and >16y. The probability of living to 80 years old, having menopause ≥ 50 years old, risk of diabetes, high blood pressure (HBP), heart failure, COPD, breast cancer, endometrial cancer, and obesity were compared among the 5 age groups of menarche. Each of the outcomes was compared between the highest and lowest education levels and 3 BMI categories (18.5-24.9, 25-29.9, and ≥ 30) among the 5 age groups (**Fig. 4A-I**). The ORs for each age group were calculated based on the coefficients (β) in logistic regression after including confounders of smoking, alcohol, education, and BMI. Compared to the group with menarche under 11 years old, which generally had the highest risk, females with menarche at 13-14y had the highest probability of living to 80 years old (OR=1.412, $P<0.001$) and having menopause ≥ 50 years old (OR=1.146, $P<0.001$), and had the lowest risk of diabetes (OR=0.773, $P<0.001$), heart failure (OR=0.835, $P<0.001$), and COPD (OR=0.728, $P<0.001$). The group with menarche at 15-16y had the lowest risks of high blood pressure (HBP) (OR=0.793, $P<0.001$) and obesity (OR=0.337, $P<0.001$). Females with menarche >16y had the lowest risks of breast cancer (OR=0.832, $P<0.05$) and endometrial cancer (OR=0.641, $P<0.05$) compared to females with menarche <11y (Table S10).

Ages at first birth were divided into 5 age groups, <21y, 21-25y, 26-30y, 31-35y, and >35y. The probability of living to 80 years old, having menopause ≥ 50 years old, risk of diabetes, AD, HBP, heart failure, cerebral infarction, digestive system disease, cervical cancer, and obesity were compared among the 5 age groups of first live birth. Compared to the group with first live birth under 21 years old, which generally had the highest risks, females with first live birth at 21-25y had the highest probability to live to 80 years old (OR=1.667, $P<0.001$), and females with first live birth at 26-30y had the highest probability to have menopause ≥ 50 years old (OR=1.184, $P<0.001$) and lowest risk of cervical cancer (OR=0.587, $P<0.01$).

Females with first live birth at 31-35y had the lowest risk of diabetes (OR=0.612, $P<0.001$). Females with first live birth >35y had the lowest risk of having AD (OR=0.474, $P<0.01$), HBP (OR=0.587, $P<0.001$), heart failure (OR=0.446, $P<0.001$), cerebral infarction (OR=0.485, $P<0.001$), digestive system disease (OR=0.569, $P<0.001$), and obesity (OR=0.416, $P<0.001$) (**Fig. 4J-S**).

Females with the number of births from 0 to 4 were included. Compared to no birth, females with 3 births had the highest probability of living to 80 years old (OR=1.818, $P<0.001$) and having menopause ≥ 50 years old (OR=1.562, $P<0.001$). Females with 4 births had the highest risk of diabetes (OR=1.399, $P<0.001$), AD (OR=1.676, $P<0.001$), HBP (OR=1.274, $P<0.001$), heart failure (OR=1.439, $P<0.001$), cerebral infarction (OR=1.470, $P<0.001$), digestive system disease (OR=1.370, $P<0.001$), and obesity (OR=1.327, $P<0.001$) but lowest risk of breast cancer (OR=0.869, $P<0.001$) (**Fig. 4T-AC**). The significant estimated effect (β) of each group compared to the youngest age groups of menarche and first live birth and 0 birth group and the ability to explain the variability of outcomes in regression analysis are exhibited in Table S10.

To estimate the combined effect of age at menarche and first live birth on diabetes, HBP, heart failure, and BMI ≥ 30 , participants were divided into 25 groups according to the menarche and first live birth ages. Females with menarche <11y and first live birth <21y had higher risks of diabetes (OR=1.881, $P<0.001$), HBP (OR= 1.436, $P<0.05$), heart failure (OR= 1.776, $P<0.01$), and BMI ≥ 30 (OR= 4.417, $P<0.001$) compared to females with menarche of 13-14y and first live birth of 26-30y (Table S11 and fig. S3).

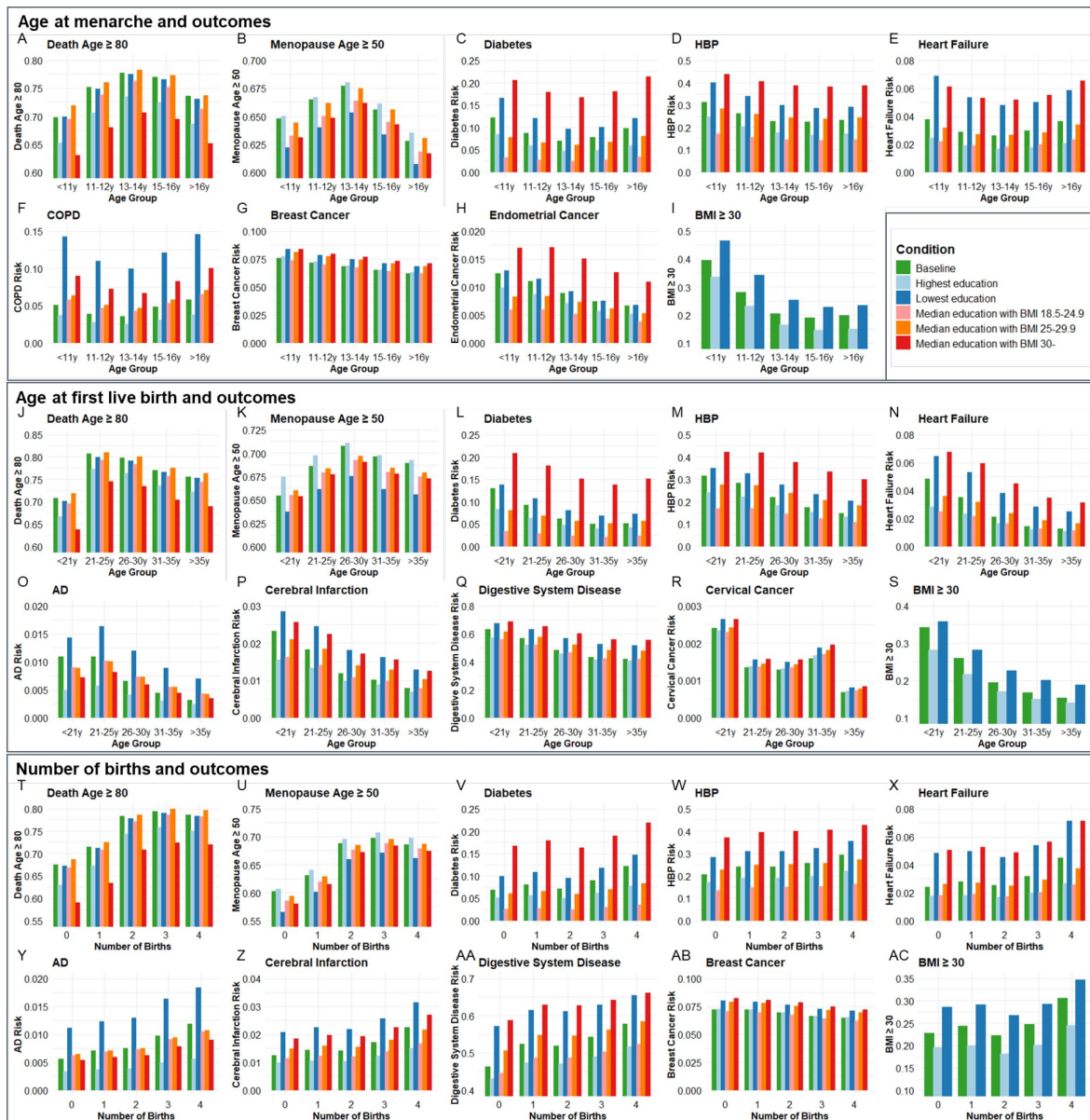


Figure 4.

The distributions of outcome age and risk according to age group of menarche and first live birth as well as number of births before and after correcting confounders.

Condition Baseline, no confounder was corrected. Conditions Highest education and Lowest education, confounders of education, smoking, and drinking were corrected, and bars were painted based on most smoking and drinking situations with the highest and lowest education levels. Conditions Median education with BMI 18.5-24.9, Median education with BMI 25-29.9, and Median education with BMI 30-, confounders of education, smoking, drinking, and BMI were corrected, and bars were painted based on median education and most smoking and drinking situation with 3 BMI categories. BMI, body mass index; HBP, high blood pressure; COPD, chronic obstructive pulmonary disease; AD, Alzheimer's disease.

The timing of reproductive milestones such as the age of menarche and age at first birth is partly governed by genetic factors, which may exert lasting impacts on an individual's health trajectory. Our study finds support for antagonistic pleiotropy by demonstrating that genetic variants that drive early menarche or early pregnancy accelerate several aging outcomes, including parental age, frailty index, and several age-related diseases such as diabetes and dementia. These results were validated in a cohort from the UK biobank, showing that women with early menarche, or early pregnancy, were associated with accelerated aging outcomes and increased risk of several age-related diseases. These results are also consistent with the disposable soma theory that suggests aging as an outcome tradeoff between an organism's investment in reproduction and somatic maintenance and repair. Consistent with a recent study suggesting the relation of early first birth with a higher likelihood of frailty²⁹, our MR analysis provided a strong association between early age of first birth and GrimAge, a DNA methylation based marker to predict epigenetic age. Our findings align with the observed negative genetic correlation between reproductive traits and lifespan that individuals with higher polygenetic scores for reproduction have lower survivorships to age 76³⁰. Prior studies linking the female reproductive factors with aging are either limited to results obtained from observational studies^{31,32} or are limited to few outcomes such as brain disorders^{33,34}. Furthermore, the origins of aging in humans have also been considered adaptive³⁵.

Using MR, we identified 128 SNPs associated with early menarche or first birth that significantly influence age-related outcomes. Among the top 25 enriched canonical signaling pathways, those involved in developmental and cellular processes are particularly significant in determining the timing of menarche and first childbirth, with nearly half of these pathways (12 out of 25) falling into this category. These developmental and cellular signaling pathways work together to regulate continuous growth and involution from puberty through menopause³⁶. Interestingly, known longevity pathways such as insulin-like growth factor 1 (IGF1), growth hormone (GH) signaling, melatonin signaling, and bone morphogenetic protein (BMP) signaling, seem to play a crucial role in regulating the timing of these reproductive events. IGF-1 signaling promotes the growth and development of reproductive organs and tissues³⁷⁻³⁹, but is also a conserved modulator of longevity⁴⁰⁻⁴³. Melatonin, known for regulating circadian rhythms, influences the timing of menarche through its effects on the hypothalamic-pituitary-gonadal axis⁴⁴ and is postulated to modulate oxidative, inflammatory, and autophagy states⁴⁵. Furthermore, different processes in early life such as cell proliferation or differentiation⁴⁶, ovarian function, and follicular development⁴⁷⁻⁴⁹ are dependent on BMP signaling. However, increased BMP signaling also contributes significantly to AD pathology⁵⁰ and impairments in neurogenesis and cognitive decline associated with aging⁵¹. Aberrant BMP signaling is also shown to be associated with age-related metabolic and cardiovascular diseases^{52,53}. Additionally, growth hormone signaling is not only vital for development during childhood and puberty⁵⁴ but also impacts longevity⁵⁵. Besides signaling pathways, genes in age at menarche and first childbirth were connected in gene-gene interaction networks. In IPA network 1, most genes were connected with either *AKT* or genes encoding for follicle-stimulating hormone (*FSH*), luteinizing hormone (*LH*), leptin receptor (*LEPR*), and inhibin subunit beta A (*INHBA*). *AKT* signaling is implicated in oocyte maturation and embryonic development⁵⁶ and mediates pregnancy-induced cardiac adaptive responses⁵⁷. *AKT* signaling also mediates age-related disease pathologies, such as AD⁵⁸. Major ovarian functions are controlled by *FSH* (follicular growth, cellular proliferation, and estrogen production) and *LH* (oocyte maturation, ovulation, and terminal differentiation of follicles) which in turn are modulated by other ovarian factors such as *INHBA* (a subunit of activin and inhibin)^{59,60}. Endocrine alterations at advanced aging or reproductive aging⁶¹ are marked by changes in *FSH* and *LH* due to altered feedback resulting from the ovarian decline in sex steroids, inhibin A, and inhibin B production⁶²⁻⁶⁴. In IPA network 2, the hub genes are connected mostly with extracellular signal-regulated kinases (*ERK*). Similar to *AKT*, *ERK* is also a member of the mitogen-activated kinase family (*MAPK*), a highly conserved signaling pathway that plays a vital role in transducing extracellular signals into a wide range of cellular responses^{65,66}, maintaining tissue integrity during aging⁶⁷, and mediating age-related disease

pathology ⁵⁸[68](#). These studies underscore the importance of considering that dysregulation or abnormal activation of any of these pathways could contribute to the onset of late-life diseases consistent with antagonistic pleiotropy in humans ²[69](#).

The thrifty gene hypothesis provides a compelling framework to explain antagonistic pleiotropy, particularly in the context of modern health challenges. The thrifty gene hypothesis suggests that genes favoring efficient energy storage were advantageous in historical periods of food scarcity, helping individuals survive through famines. However, in today's environment of abundant food, the same genes contribute to obesity and metabolic diseases like diabetes. Similarly, in the context of antagonistic pleiotropy, we hypothesize that genes that enhance early-life reproductive success favor efficient energy storage to support reproductive health. This is supported by our observation that early pregnancy is associated with increasing BMI, a recognized key factor in systemic aging ⁷⁰[71](#). In support of this, we found a lack of significant association between early menarche or age at first birth for CHF and cervical cancer after removal of BMI as a confounder in MR analysis. Based on regression analysis at the population level (**Fig. 4** [4](#)), higher BMI was associated with higher risks of cardiovascular diseases and diabetes. Thus, an increase in BMI is one of the factors that explains the increased risk of age-related disease seen due to the exposure of early pregnancy or the prevalence of genes that enhance early reproductive success. These results are also consistent with calorie restriction being a robust way to extend healthspan and lifespan in many species ⁷²[73](#).

Several limitations of our study need to be considered. Firstly, all the associations are explored at the genetic level with MR and population level with the UK Biobank. However, to confirm causal relationships, further validation through *in vitro* and *in vivo* research is essential. Secondly, though we excluded confounder-related SNPs and addressed potential pleiotropic effects, there may also be potential bias which could be addressed by extending these findings to other populations. Thirdly, significant heterogeneity and pleiotropy remained in some association analyses, which need further exploration. Given the increasing age of first childbirth in modern times, the ideal period of first childbirth for both slowing down female aging and benefiting fetal development needs more research ⁷³[74](#). In summary, our study underscores the complex relationship between genetic legacies and modern diseases. Understanding these genetic predispositions can inform public health strategies and their relevance to age-related disease outcomes in women of various ethnic groups. For example, interventions could be tailored not only to mitigate the risks associated with early reproductive timing but also to address lifestyle factors exacerbating conditions linked to thrifty genes, like diet and physical activity.

Methods

All the MR research data is from the public genome-wide association studies (GWAS) databases of the IEU Open GWAS project (<https://gwas.mrcieu.ac.uk/> [4](#)) and PubMed. The target analysis is based on public datasets. The population data is from the UK Biobank. No definite personal information was included. No additional ethical approval was required for our study.

Exposure data

2 traits were included as female reproductive activity exposures, involving age at menarche ⁷[75](#) (GWAS ID: ebi-a-GCST90029036) and age at first birth ⁸[8](#) (GWAS ID: ebi-a-GCST90000048).

Outcome data

The aging outcomes involved general aging, organ aging/disease, and organ cancers (Table S1). Frailty index ⁷⁴[74](#) (GWAS ID: ebi-a-GCST90020053), father's age at death ⁷⁵[75](#) (GWAS ID: ebi-a-GCST006700), mother's age at death ⁷⁵[75](#) (GWAS ID: ebi-a-GCST006699), and DNA methylation

GrimAge acceleration⁷⁶ (GWAS ID: ebi-a-GCST90014294) were included as general aging outcomes. Specific aging diseases and aging levels included age at menopause onset⁷⁷ (GWAS ID: ebi-a-GCST90029037), late-onset Alzheimer's disease⁷⁷ (LOAD) (PMID: 30820047), osteoporosis⁷⁸ (GWAS ID: ebi-a-GCST90038656), and type 2 diabetes⁷⁹ (GWAS ID: ebi-a-GCST90018926), chronic heart failure⁷⁹ (CHF) (GWAS ID: ebi-a-GCST90018806), essential hypertension (GWAS ID: ukb-b-12493), facial aging (GWAS ID: ukb-b-2148), eye aging⁸⁰ (PMID: 36975205), cirrhosis (GWAS ID: finn-b-CIRRHOSIS_BROAD), chronic kidney disease⁸¹ (CKD) (GWAS ID: ebi-a-GCST008026), early onset chronic obstructive pulmonary disease (COPD) (GWAS ID: finn-b-COPD_EARLY), and gastrointestinal or abdominal disease (GAD)⁷⁸ (GWAS ID: ebi-a-GCST90038597). Organ cancers included breast cancer⁷⁹ (GWAS ID: ebi-a-GCST90018799), ovarian cancer⁷⁹ (GWAS ID: ebi-a-GCST90018888), endometrial cancer⁸² (GWAS ID: ebi-a-GCST006464), and cervical cancer (GWAS ID: ukb-b-8777).

SNPs selection

We identified single nucleotide polymorphisms (SNPs) associated with exposure datasets with $p < 5 \times 10^{-8}$ ^{83,84}. In this case, 249 SNPs and 67 SNPs were selected as eligible instrumental variables (IVs) for exposures of age at menarche and age at first birth, respectively. All selected SNPs for every exposure would be clumped to avoid the linkage disequilibrium ($r^2 = 0.001$ and $kb = 10,000$). Then we identified whether there were potential confounders of IVs associated with the outcomes based on a database of human genotype-phenotype associations, PhenoScanner V2^{85,86} (<http://www.phenoscanter.medschl.cam.ac.uk/>), with a threshold of $p < 1 \times 10^{-5}$. IVs associated with education, smoking, alcohol, activity, and other confounders related to outcomes would be excluded. During the harmonization process, we aligned the alleles to the human genome reference sequence and removed incompatible SNPs. Subsequent analyses were based on the merged exposure-outcome dataset. We calculated the F statistics to quantify the strength of IVs for each exposure with a threshold of $F > 10$ ⁸⁷. If the effect allele frequency (EAF) was missing in the primary dataset, EAF would be collected from dsSNP (<https://www.ncbi.nlm.nih.gov/snp/>) based on the population to calculate the F value.

MR analysis

All analysis was performed using R software (version 4.3.1, R Foundation for Statistical Computing, Vienna, Austria) and RStudio software (version 2023.09.0 Posit, PBC, Boston, USA) with TwoSampleMR (version 0.5.7) and MRPRESSO (version 1.0) packages.

Five two-sample MR methods were applied for analysis, including inverse variance weighted model (IVW), MR Egger regression model (MER), weighted median model (WMM), simple mode, and weighted mode. A pleiotropy test was used to check if the IVs influence the outcome through pathways other than the exposure of interest. A heterogeneity test was applied to ensure whether there is a variation in the causal effect estimates across different IVs. Significant heterogeneity test results indicate that some instruments are invalid or that the causal effect varies depending on the IVs used. MRPRESSO was applied to detect and correct potential outliers of IVs with NbDistribution = 10,000 and threshold $p = 0.05$. Outliers would be excluded for repeated analysis. The causal estimates were given as odds ratios (ORs) and 95% confidence intervals (CI). Leave-one-out analysis was conducted to ensure the robustness of the results by sequentially excluding each IV and confirming the direction and statistical significance of the remained SNPs.

In two-step MR, we mainly use IVW to access the causal associations between exposure and outcome, exposure and mediator, and mediator and outcome. Body mass index (BMI) (GWAS ID: ukb-b-19953) was applied as a mediator. Similar steps of two-sample MR were repeated in the analysis after excluding confounders. The mediator effect (ME) and direct effect (DE) were calculated. The MEs were calculated with the formula: $\beta_1 \times \beta_2$; DEs were calculated

according to the formula: $\beta = (\beta_1 \times \beta_2)$, β stands for the total effect obtained from the primary analysis, β_1 stands for the effect of exposure on the mediator, and β_2 stands for the effect of the mediator on the outcome.

Colocalization analysis

To improve the robustness of our research, colocalization analysis was conducted with packages *gwasglue* (version 0.0.0.9000), *coloc* (version 5.2.3), and *gassocplot* (version 1.0) between the exposures and outcomes revealing significant associations after BMI-related SNPs were excluded. SNPs with significant P values in single SNP OR analysis were set as target SNPs. SNP-level colocalization was conducted with 50kb ⁸⁸ windows around each target SNP. EAF was set to 0.5 if it was missed in the primary datasets. The Bayesian algorithm in the *coloc* package generates posterior probabilities (PP) for the hypothesis that both traits are associated and share the same single causal variant at a specific locus (H_4) ⁸⁹. SNP with significant results in the colocalization analysis would be removed and two-sample MR analyses would be conducted again.

Target Analysis

We employed the Ingenuity Pathway Analysis (IPA) software (version 01-22-01; Ingenuity Systems; QIAGEN) to investigate various gene-related aspects associated with age at first birth and menarche. IPA is a widely used bioinformatics tool for interpreting high-throughput data ⁹⁰. Briefly, the SNPs/genes from MR analysis were uploaded into Qiagen's IPA system for core analysis and then the outcome was overlaid with the global molecular network in the Ingenuity pathway knowledge base (IPKB). IPA was performed to identify canonical pathways, diseases, and functions, and to investigate gene networks. Additionally, the Chemical Biology Database (ChEMBL) (<https://www.ebi.ac.uk/chembl/> ²⁷), a large-scale database of bioactive molecules for drug discovery ²⁷, and the DrugBank database ²⁸, a comprehensive database on FDA-approved drugs, drug targets, mechanisms of action, and interactions (<https://go.drugbank.com/> ²⁸), were used to identify candidate genes targeted by approved drugs in clinical trials or under current development phases.

Population validation

The MR results were further validated based on the UK Biobank with package *stats* (version 4.3.1). In addition to age at menarche and first live birth, the number of births is listed as an independent variable for the validation analysis. Based on the significant associations, regression analysis was adopted to explore the effect of independent variables and related confounders on outcomes. Ages at menarche were divided into 5 age groups, <11y, 11-12y, 13-14y, 15-16y, and >16y. Ages at first live birth were divided into 5 age groups, <21y, 21-25y, 26-30y, 31-35y, and >35y. Females with the number of births from 0 to 4 were included. Education was divided into 7 levels based on the degrees. Smoking was divided into 2 categories, ever smoking or not. Drinking was divided into 3 categories, never, previous, and current drinking. BMI was divided into four categories, <18.5, 18.5-24.9, 25-29.9, and ≥ 30 based on the average value of 4 records. The outcome results were preprocessed based on the baseline information collection, follow-up outcome, surgical history, and ICD-10 diagnosis summary. Logistic regression was used for the analysis of disease risks (categorical variables). The coefficients (β) in logistic regression were log-odds ratios. Although the continuous variables exhibited mild to moderate skewed distributions, considering the large sample size, linear regression was used for the analysis of age at death, menopause age, and BMI in the first step. Then logistic regression was applied to confirm the robustness of the results with the outcome of death age ≥ 80 , menopause age ≥ 50 , and BMI ≥ 30 . Confounders for regression analysis I include education, smoking, and drinking. Confounders for regression analysis II include education, smoking, drinking, and BMI. Variance Inflation Factors (VIF) were calculated for each predictor in regression analysis to avoid multicollinearity. Values of VIF were less than 1.5 for each predictor. For outcomes of diabetes, HBP, heart failure, and BMI ≥ 30 , the combined effect of age at menarche and first birth was analyzed. 179,821 participants were divided into 25 groups

according to the menarche and first live birth age groups. All other groups were compared to the group with menarche of “13-14y” and the first birth of “26-30y” in logistic regression (confounders including education, smoking, drinking, and BMI for diabetes, HBP, and heart failure; including education, smoking, and drinking for BMI $\geq$ 30).

Data Availability

All data are available in the main text, supplementary materials, or public database. The population data is available through the UK Biobank.

Data and materials availability

All data are available in the main text, supplementary materials, or public database. The population data is available through the UK Biobank.

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Author contributions

Conceptualization: YX, VT, PS, LLF, and PK. MR analysis and UK biobank validation: YX. Gene target analysis: VT. Funding acquisition: PK and PS. Writing original draft: YX, PS, and VT. All authors reviewed and edited the manuscript.

Competing interests

Authors declare that they have no competing interests.

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Reviewer #1 (Public review):

Summary:

The present study aims to associate reproduction with age-related disease as support of the antagonistic pleiotropy hypothesis of ageing, predominantly using Mendelian Randomization. The authors found evidence that early-life reproductive success is associated with advanced ageing.

Strengths:

Large sample size. Many analyses.

Weaknesses:

There are some errors in the methodology, that require revisions.

In particular, the main conclusions drawn by the authors refer to the Mendelian Randomization analyses. However, the authors made a few errors here that need to be reconsidered:

(1) Many of the outcomes investigated by the authors are continuous outcomes, while the authors report odds ratios. This is not correct and should be revised.

(2) Some of the odds ratios (for example the one for osteoporosis) are really small, while still reaching the level of statistical significance. After some checking, I found the GWAS data used to generate these MR estimates were processed by the program BOLT-LLM. This program is a linear mixed model program, which requires the transformation of the beta estimates to be useful for dichotomous outcomes. The authors should check the manual of BOLT-LLM and recalculate the beta estimates of the SNP-outcome associations prior to the Mendelian Randomization analyses. This should be checked for all outcomes as it doesn't apply to all.

(3) The authors should follow the MR-Strobe guidelines for presentation.

(4) The authors should report data in the text with a 95% confidence interval.

(5) The authors should consider correction for multiple testing.

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Reviewer #2 (Public review):

Summary:

The authors present an interesting paper where they test the antagonistic pleiotropy theory. Based on this theory they hypothesize that genetic variants associated with later onset of age at menarche and age at first birth have a positive causal effect on a multitude of health outcomes later in life, such as epigenetic aging and prevalence of chronic diseases. Using a mendelian randomization and colocalization approach, the authors show that SNPs associated with later age at menarche are associated with delayed aging measurements, such as slower epigenetic aging and reduced facial aging, and a lower risk of chronic diseases, such as type 2 diabetes and hypertension. Moreover, they identified 128 fertility-related SNPs that are associated with age-related outcomes and they identified BMI as a mediating factor for disease risk, discussing this finding in the context of evolutionary theory.

Strengths:

The major strength of this manuscript is that it addresses the antagonistic pleiotropy theory in aging. Aging theories are not frequently empirically tested although this is highly necessary. The work is therefore relevant for the aging field as well as beyond this field, as the antagonistic pleiotropy theory addresses the link between fitness (early life health and reproduction) and aging.

Points that have to be clarified/addressed:

(1) The antagonistic pleiotropy is an evolutionary theory pointing to the possibility that mutations that are beneficial for fitness (early life health and reproduction) may be detrimental later in life. As it concerns an evolutionary process and the authors focus on contemporary data from a single generation, more context is necessary on how this theory is accurately testable. For example, why and how much natural variation is there for fitness outcomes in humans? How do genetic risk score distributions of the exposure data look like? Also, how can the authors distinguish in their data between the antagonistic pleiotropy theory and the disposable soma theory, which considers a trade-off between investment in reproduction and somatic maintenance and can be used to derive similar hypotheses? There is just a very brief mention of the disposable soma theory in lines 196-198.

(2) The antagonistic pleiotropy theory, used to derive the hypothesis, does not necessarily distinguish between male and female fitness. Would the authors expect that their results extrapolate to males as well? And can they test that?

(3) There is no statistical analyses section providing the exact equations that are tested. Hence it's not clear how many tests were performed and if correction for multiple testing is necessary. It is also not clear what type of analyses have been done and why they have been done. For example in the section starting at line 47, Odds Ratios are presented, indicating that logistic regression analyses have been performed. As it's not clear how the outcomes are defined (genotype or phenotype, cross-sectional or longitudinal, etc.) it's also not clear why logistic regression analysis was used for the analyses.

(4) Mendelian Randomization is an important part of the analyses done in the manuscript. It is not clear to what extent the MR assumptions are met, how the assumptions were tested, and if/what sensitivity analyses are performed; e.g. reverse MR, biological knowledge of the studied traits, etc. Can the authors explain to what extent the genetic instruments represent their targets (applicable expression/protein levels) well?

(5) It is not clear what reference genome is used and if or what imputation panel is used. It is also not clear what QC steps are applied to the genotype data in order to construct the genetic instruments of MR.

(6) A code availability statement is missing. It is understandable that data cannot always be shared, but code should be openly accessible.

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